Bed Model

Software Requirements / Logical Data Model

1. Purpose

Represents an individual bed or accommodation slot that can be tracked and potentially allotted to a student.

2. Responsibilities

• Uniquely identify each bed.

• Associate a bed with its room and hostel through the room relationship.

• Support availability checks and prevention of duplicate active allotments.

3. Proposed Logical Attributes

The following is a documentation proposal, not a claim that every field is already implemented. Confirm exact field names, types, constraints and cardinalities against the project ERD/database schema.

Attribute	Meaning / Requirement
bed_id	Unique bed identifier.
room_id	Reference to the room containing the bed.
bed_number_or_code	Bed label within the room.
status	Configured state such as available, occupied or unavailable, if these states are stored.
notes	Optional administrative notes, if required.

4. Relationships

• Each bed should belong to one room at a time.

• A bed may have multiple historical allotment records but should not have conflicting active allotments.

• Availability is relevant to application review, allotment and reporting.

5. Validation and Business Rules

• Do not create two simultaneous active allotments for the same bed.

• Check bed availability at the time of allotment, not only when the application is first opened.

• Changes to bed status must be restricted to authorized users.

• Historical allotments should be retained according to the project’s record-retention policy.

6. Traceability / Notes

The project objectives explicitly include tracking room/bed availability and preventing duplicate bed allotments.

Status: Draft for review. Align this document with the approved SRS, ERD, class diagram, and institutional policies before treating it as final.